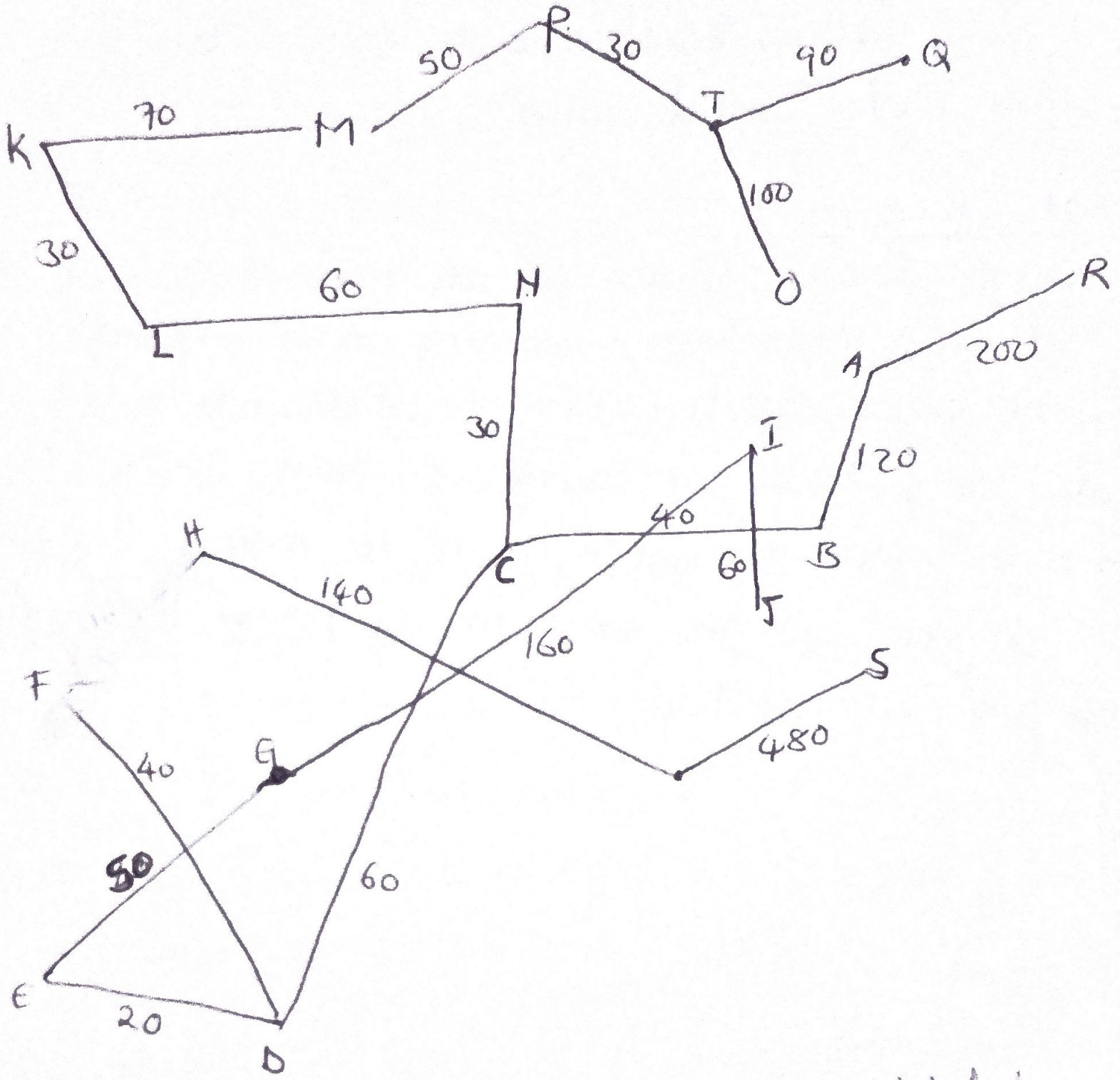
Question 6

. Yes A Minimum spanning tree exist for a graph is connected, undirected and weighted
The spanning tree is shown below.



The graph has all three properties. It is weighted because every road has a distance, undirected because roads can be travelled in both directions, and connected because every one of the 20 ~~nodes can~~ nodes can be reached from every other
This is confirmed by the program: Kruskal's algorithm selected 19 edges, which is $N-1$ for $N=20$ nodes so no node was left disconnected.